

KOMATSU AB, Japan

WMO: 477040

Lat: 36.394N

Lon: 136.407E

Elev: 10

StdP: 101.21

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-1.0	-7.9	1.9	3.2	-6.2	2.2	2.9	15.1	3.5	13.4	3.1	2.8	120	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	33.2	24.5	32.1	24.5	30.9	24.3	26.1	30.7	25.5	30.0	24.9	29.2	5.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	19.9	28.6	24.1	19.0	28.0	23.2	18.0	27.5	80.6	30.8	78.2	30.5	75.4	29.3	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.8	8.5	-4.2	36.2	1.7	1.3	-5.5	37.1	-6.4	37.8	-7.4	38.5	-8.6	39.5		
			-5.1	27.3	1.5	1.3	-6.1	28.2	-7.0	28.9	-7.8	29.7	-8.9	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	3.7	4.0	7.1	12.1	17.3	21.2	25.7	27.1	22.9	17.3	11.4	6.4
	DBStd	8.53	2.39	2.75	3.26	3.57	2.83	2.48	2.53	2.18	2.83	2.90	3.06	2.84
	HDD10.0	621	194	170	103	18	0	0	0	0	0	19	116	
	HDD18.3	2083	453	401	349	189	53	4	0	2	55	208	369	
	CDD10.0	2355	1	2	13	81	227	336	487	531	387	225	61	5
	CDD18.3	776	0	0	0	2	22	89	229	273	138	22	1	0
	CDH23.3	6339	0	0	1	21	126	382	1992	2845	885	86	2	0
	CDH26.7	2118	0	0	0	3	16	70	677	1113	228	12	0	0
Wind	WSAvg	3.5	4.4	4.0	4.0	3.7	3.3	2.9	3.0	3.1	3.2	3.6	4.3	
Precipitation	PrecAvg	2267	272	164	155	133	143	158	228	151	207	155	211	292
	PrecMax	2718	456	256	232	220	356	283	577	379	354	247	414	646
	PrecMin	1543	100	54	70	47	60	45	53	28	69	82	122	125
	PrecStd	272	87	50	44	47	73	58	129	83	96	44	56	120
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.9	21.1	26.0	28.8	31.0	35.0	35.9	33.2	28.2	22.8	17.9
		MCWB	9.9	11.9	12.1	15.9	18.2	22.4	24.2	24.8	23.3	20.4	15.9	12.2
	2%	DB	11.2	13.2	18.1	23.0	26.1	28.8	33.0	34.0	30.9	25.2	20.2	14.9
		MCWB	6.9	8.6	10.7	14.2	16.8	21.3	24.4	24.8	23.3	19.1	14.7	10.4
	5%	DB	9.2	11.0	15.8	20.9	24.8	26.9	31.8	32.8	29.1	23.9	18.9	13.0
		MCWB	5.9	6.8	9.9	13.5	16.8	20.9	24.4	24.8	22.9	18.3	14.0	9.4
	10%	DB	8.0	9.0	13.2	18.9	23.0	25.2	30.1	31.2	27.8	22.2	17.0	11.2
		MCWB	5.2	5.8	8.7	12.6	16.1	20.3	23.8	24.7	22.2	17.2	13.1	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.6	13.8	17.7	20.4	23.9	26.4	27.0	25.4	22.0	17.4	13.6
		MCDB	13.7	17.3	17.9	22.5	25.3	27.8	31.1	31.3	30.0	26.2	20.2	16.7
	2%	WB	7.8	9.7	12.0	15.6	18.8	22.7	25.6	26.4	24.5	20.6	15.9	11.2
		MCDB	10.1	12.6	16.4	20.3	23.4	26.0	30.2	30.9	28.4	23.8	19.1	13.9
	5%	WB	6.3	7.5	10.6	14.4	17.9	22.0	25.0	25.9	23.7	19.4	14.7	9.9
		MCDB	8.6	10.0	14.2	19.0	22.4	25.4	29.4	30.5	27.4	22.7	17.9	12.3
	10%	WB	5.2	5.8	9.3	13.2	17.0	21.2	24.5	25.3	22.9	18.3	13.5	8.6
		MCDB	7.6	8.4	12.9	17.4	21.2	24.3	28.7	29.8	26.6	21.6	16.5	11.1
Mean Daily Temperature Range	5% DB	MDBR	6.1	7.0	8.7	9.4	8.7	6.7	6.4	7.4	7.5	8.5	8.4	7.1
		MCDBR	9.0	11.0	13.2	13.1	12.0	9.2	8.8	9.2	8.9	9.9	11.1	9.9
		MCWBR	6.2	7.7	8.2	6.7	5.7	4.2	3.4	3.7	4.1	5.2	6.8	6.8
	5% WB	MCDBR	8.3	10.0	10.8	10.3	9.2	6.7	7.3	7.7	7.6	8.1	9.2	8.9
		MCWBR	6.3	7.7	7.6	6.0	5.2	3.5	3.4	3.7	4.0	4.8	6.4	6.6
Clear-Sky Solar Irradiance	taub	0.439	0.468	0.533	0.556	0.556	0.575	0.568	0.541	0.489	0.456	0.440	0.440	
	taud	1.951	1.897	1.802	1.812	1.874	1.908	1.988	2.067	2.166	2.141	2.092	1.985	
	Ebn at Noon	704	741	732	744	752	736	738	749	768	751	707	672	
	Edh at Noon	141	168	202	210	201	194	178	161	139	130	122	129	
All-Sky Solar Radiation	RadAvg	1.60	2.37	3.61	4.62	5.23	5.02	4.88	5.12	3.94	3.13	2.07	1.41	
	RadStd	0.15	0.24	0.31	0.38	0.57	0.41	0.71	0.54	0.31	0.25	0.18	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page